

Consumer Valuation of Battery Health Attributes in Used Battery Electric Vehicle Markets: A Discrete Choice Experiment

Xiatian Iogansen^{a,b,1}, Zain Hoda^{b,2}, Christina Gore^{a,*,3}, Joshua D. Kneifel^{a,4},
Sindhu Ranganath^{a,b,5} and John P. Helveston^{b,6}

^aApplied Economics Office, Engineering Laboratory, National Institute of Standards and Technology, Maryland, USA

^bDepartment of Engineering Management and Systems Engineering, George Washington University, Washington D.C., USA

1. Introduction

The used vehicle market is central to U.S. automobile ownership. Used vehicle transactions account for roughly 70% of all passenger vehicle purchases and are projected to grow steadily through the coming decade (Market Research Future, 2025). An increasing number of used battery electric vehicles (BEVs) are entering the secondary market, and demand is accelerating: sales rose 34.2% year-over-year through early 2025, while average days of supply fell by 21.5% (Cox Automotive Inc., 2025). Despite this momentum, consumer hesitation toward used BEVs remains pronounced. Fernandes (2023) reports that 52% of U.S. adults are unlikely to consider purchasing a used BEV, compared to 28% who express openness to the idea. Consumers most often cite *uncertainty about vehicle and battery condition* (41%) and *concerns about reliability relative to new EVs* (23%) as primary barriers. This gap between perceived and actual risk points to an information problem: buyers lack information about battery health at the point of sale. This is a considerable risk, as the battery can account for up to 30% of a used BEV's value (Boudway, 2020).

Unlike conventional vehicles, where odometer readings and vehicle age are reasonable proxies for overall wear, unique attributes of used BEVs, including battery degradation, battery refurbishment, and warranty coverage, increase the uncertainty for consumers. This paper asks how consumers value battery health attributes among used BEVs and how preferences vary across consumer segments. We address these questions using a discrete choice experiment (DCE) in an online survey administered to a national U.S. sample of 3,072 adults who reported an intention to purchase a used car or SUV within the next two years. The DCE presents respondents with six repeated choices among hypothetical used BEVs that vary in mileage, purchase price, electric range, battery degradation rate, and battery refurbishment history.

We estimate a latent class choice model (LCCM) which reveals six distinct consumer segments that differ markedly in their sensitivity to battery quality signals, price, and the no-choice option. Vehicles with documented performance indicators and battery health attributes represent a material pricing premium opportunity with the majority of the market. To our knowledge, this study provides the first WTP estimates for battery health attributes in the U.S. used BEV market. These findings carry direct implications for used vehicle dealers, OEMs, and certification service providers seeking to develop market infrastructure for used BEVs.

2. Survey Design, Data Collection and Modeling Approach

Each respondent completes six repeated choice tasks. Prior to the choice tasks, respondents are shown descriptions of all attributes (see Table 1). During the DCE design phase, we used range at the time of manufacture (Year 0) and an annual degradation rate to characterize battery condition. To improve comprehension for respondents, we transformed these underlying design parameters and presented range and SOH at Year 3 (the current purchase date) and Year 8 (five

*Corresponding author

¹xiatian.iogansen@gwu.edu & xiatian.iogansen@nist.gov, <https://orcid.org/0000-0002-4851-1323>

²zain.hoda@gwu.edu, <https://orcid.org/0009-0003-0958-1868>

³christina.gore@nist.gov, <https://orcid.org/0000-0002-3586-6918>

⁴joshua.kneifel@nist.gov, <https://orcid.org/0000-0002-3114-5531>

⁵sindhu.ranganath@nist.gov, <https://orcid.org/0000-0001-5764-9773>

⁶jph@gwu.edu, <https://orcid.org/0000-0002-2657-9191>

1 years after purchase).Figure 1 shows an example choice task, in which respondents select among three used BEVs or
2 a “None of the above” option.

3 In addition to the DCE, the survey collects information across three broader topic areas, including their current
4 household vehicle fleet and their future vehicle purchasing intention, their knowledge, attitudes, and perceptions
5 regarding BEVs and battery technology, as well as demographic information.

Table 1
BEV choice experiment attributes and levels.

Attribute	Definition	Levels			
		Low-budget (≤\$20K) Car	High-budget (>\$20K) Car	Low-budget (≤\$20K) SUV	High-budget (>\$20K) SUV
Mileage	The total number of miles a vehicle has traveled since manufacture.	15,000–60,000 miles, in increments of 500 miles.			
Battery refurbishment history	The repair or replacement work that has been performed on the vehicle's main battery pack.	(1) <i>Original</i> : The battery remains in its factory-original condition with no repairs or replacements. (2) <i>Some battery cells replaced</i> : A portion of the battery cells has been replaced to address performance issues or extend battery life; the remaining cells are original. (3) <i>Entire battery pack replaced</i> : The full battery pack has been replaced with a new, refurbished, or used pack, typically after battery performance declined below a specified threshold.			
Electric range at delivery (Year 0)	The maximum distance the vehicle can travel on a full battery charge under typical driving conditions, as rated at the time of manufacture.	50–150 miles (50-mile increments)	100–250 miles (50-mile increments)	150–250 miles (50-mile increments)	200–350 miles (50-mile increments)
Battery annual degradation rate and state of health (SOH)	The average annual percentage loss in electric range and battery SOH. The remaining range and SOH at Year 3 (current) and Year 8 (five years from purchase) are derived from this rate.	1%–8%, in increments of 1%.			
Purchase price	The total cost of the vehicle in dollars, including down payment, monthly payments, taxes, and fees.	\$10,000–\$20,000 (\$2,000 increments)	\$20,000–\$40,000 (\$5,000 increments)	\$15,000–\$25,000 (\$2,000 increments)	\$25,000–\$45,000 (\$5,000 increments)

Assume you are able to purchase a 3-year-old used vehicle that looks like the one you selected (see photo). Which of the following versions of that vehicle would you be most likely to purchase? Please compare the key features shown for each option before making your selection.



If these were your only options, which would you choose?

Option 1	Option 2	Option 3	Option 4
Mileage: 60,000	Mileage: 30,000	Mileage: 45,000	
Battery condition: Refurbished Cell-Replaced	Battery condition: Original Battery	Battery condition: Refurbished Cell-Replaced	
Range on a full charge: Current: 175 miles (Battery Health: 88%)	Range on a full charge: Current: 275 miles (Battery Health: 78%)	Range on a full charge: Current: 185 miles (Battery Health: 91%)	
Expected in 5 years: 145 miles (Battery Health: 72%)	Expected in 5 years: 180 miles (Battery Health: 51%)	Expected in 5 years: 155 miles (Battery Health: 78%)	
Purchase price: \$ 30,000	Purchase price: \$ 25,000	Purchase price: \$ 35,000	
			Even if these were my best options, I would not choose any of these vehicles

Figure 1: Example of a choice experiment.

To identify discrete consumer segments that differ systematically in their battery quality sensitivity and price responsiveness, we estimate a Latent Class Choice Model (LCCM) on 2,916 respondents (17,496 choice observations) after excluding cases with missing covariate data. The LCCM assumes that individuals can be assigned to a finite number of unobserved groups (classes) within which preferences are homogeneous, but across which they differ. The LCCM consists of two simultaneously estimated components (Figure 2). The *class membership model* assigns each respondent probabilistically to a latent class as a function of observable individual characteristics. The *class-specific choice model* then estimates the conditional probability of each vehicle choice within each class. In addition, a set of *inactive indicators*—attitudinal and behavioral variables excluded from estimation due to potential collinearity or theoretical considerations—are used to further characterize the classes descriptively after estimation. The formal model structure follows the notation of Chen et al. (2023).

A key output of the LCCM is class-specific WTP for each BEV attribute. Under the assumption that the marginal utility of price is negative and constant across alternatives, WTP for a given attribute in a given class is computed as the negative ratio of the attribute coefficient to the price coefficient. This represents the maximum dollar amount a representative respondent in that class would pay, or accept as compensation, for a one-unit change in the attribute, holding all other attributes constant.

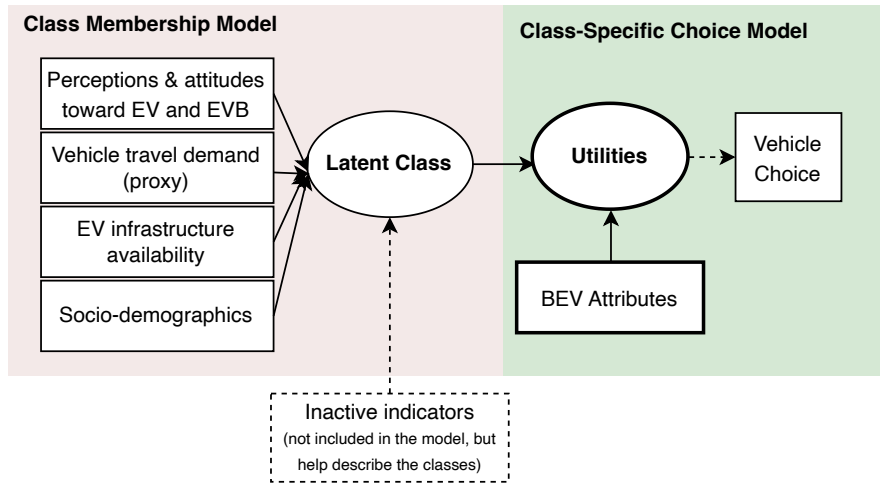


Figure 2: LCCM model framework.

3. Results and Discussion

Class 1: Battery Health Attentives ($n=661$, 22.7%).

Class 1 is the largest class, uniquely defined by its battery health sensitivity. Range loss aversion is the highest across classes, at \$2,310*** per percentage point in the 5–12% degradation segment, \$2,330*** in the 12–24% segment, and \$2,140 in the 24%+ segment. Both refurbishment types also generate meaningful disutility (-\$2,740* for pack replacement; -\$3,760** for cell replacement). Their range WTP is consistent across the three piecewise segments (\$4,570, \$4,610, and \$5,310**), indicating a steady marginal valuation of additional range without a strong threshold effect, though only the highest tier reaches statistical significance.

Members of Class 1 have an average household income of \$91,340 and are relatively young (36.5 years). They demonstrate above-average BEV knowledge (76.2%), moderate electrical outlet access (46.0%), and a high rate of EV-owning neighbors (43.5%). These well-informed consumers are not necessarily purchasing range; rather, they are purchasing battery reliability at the point of sale and protection against future performance loss.

Class 2: Multi-Attribute Maximizers ($n=606$, 20.8%).

Class 2 exhibits statistically significant valuations across all vehicle attributes. This class places a high premium on crossing the minimum viable range threshold, with still-substantial but diminishing WTP at higher range levels (\$11,830***, \$9,450***, \$9,500***). Range loss aversion is significant across all three degradation segments (-\$780***, -\$760***, -\$390* per percentage point). Refurbishment disutility is among the highest of the attribute-sensitive classes (-\$5,760*** for pack replacement; -\$6,070*** for cell replacement).

What further distinguishes Class 2 is its opt-out pattern: only 5.2% of members never opt out. Class 2 members exhibit high range anxiety (83.4%) and are particularly averse to battery refurbishment (only 18.8% disagree that refurbished batteries are functionally inferior). They also report average household incomes (\$86,920) and lower risk tolerance. These characteristics lead them to evaluate all vehicle attributes cautiously and at once. When a choice set presents no option that reaches a minimal quality threshold across all dimensions at once, respondents in this class choose to opt out rather than settle for the least-bad alternative.

Class 3: Range-Focused, EV-Knowledgeable Consumers ($n=485$, 16.6%).

Class 3 is distinguished by an exceptionally high WTP for range: \$38,700*** per 100 miles for vehicles with 40–130 miles of range, declining to \$20,820*** (130–200 miles) and \$25,470*** (200+ miles). Their range loss aversion is moderate and uniformly significant across all degradation segments (-\$870***, -\$690***, -\$470***). WTP for refurbishment is negligible and statistically insignificant, suggesting that this class focuses more on current and future performance than on battery refurbishment history.

Class 3 represents the most EV-knowledgeable segment in the sample. They hold the most favorable attitudes toward refurbished batteries (65.5% agree they are environmentally beneficial; 35.5% disagree that they are functionally inferior). Their households show the lowest rate of ICEV-only composition (74.1%), and they are the most likely class to prefer a car over an SUV (57.2%). They have the highest electrical outlet access (57.5%), the strongest BEV knowledge (81.1%), and the highest subsidy awareness (31.9%). They also have the highest household income (\$97,120), the highest employment rate (71.5%), and are the most risk-tolerant of all classes (39.0% agree with a risk-taking characterization).

Class 4: Budget-constrained, Low-WTP Consumers ($n=373$, 12.8%).

Class 4 has the largest price coefficient in absolute value (-4.017^{***}), suggesting they are budget-constrained buyers. Range WTPs are modest but statistically significant at lower range segments (\$2,810** for 40–130 miles; \$1,800* for 130–200 miles), and range loss aversion is marginal ($-\$170^{**}$ and $-\$170^{***}$ in the two upper degradation segments). WTP for refurbishment is negligible and statistically insignificant, suggesting that battery service history is largely immaterial in this group's purchasing calculus.

Class 4 members have the lowest average household income (\$70,000) and the lowest next-vehicle budget (\$19,660). They have the lowest homeownership rate (43.0%) and the highest share of renters (48.8%), the fewest household vehicles (1.9 on average), and the highest share of ICEV-only households (87.1%). Despite their strong price sensitivity, 91.2% of Class 4 members never select the no-choice option, indicating their interest in used vehicles, but their WTP is dominated by price rather than battery quality.

Class 5: Opt-Out Dominant, BEV-Skeptical Consumers ($n=252$, 8.7%).

Class 5 is opt-out dominant: 75.4% of members selected the no-choice option in all six tasks. As a result, most WTP estimates for vehicle attributes are statistically indistinguishable from zero, with the exception of a moderate disutility for range loss in the 5–12% segment ($-\$1,620^{*}$ per percentage point) and significant penalties for both pack replacement ($-\$6,350^{*}$) and cell replacement ($-\$7,320^{*}$).

Class 5 represents consumers who are not yet viable participants in the used BEV market. The majority (70.4%) strongly disagree that they are likely to purchase a used BEV. They are the oldest class on average (47.1 years), the least risk-tolerant (24.0%), and the least environmentally aware. Their household fleets are almost exclusively internal combustion engine vehicles (ICEVs), with 89.3% of households reporting no BEVs and 93.3% driving an ICEV as their primary vehicle. BEV-related knowledge is low (68.5% correctly answer the BEV knowledge question; only 10.6% are aware of federal BEV subsidies), while range anxiety is the highest of any class (87.2%). Attitudes toward refurbished batteries are notably negative. Infrastructure readiness is also limited: only 32.8% have home electrical outlet access, and only 24.7% report a neighbor who owns or leases a BEV or PHEV.

Class 6: Low-Engagement, Attribute Non-Attendance Respondents ($n=539$, 18.5%).

The most prominent feature of this class is that it has the price coefficient closest to zero (-0.144^{**}), which mechanically inflates all estimated WTP ratios. The class also shows the pattern that the choice-modeling literature associates with attribute non-attendance: one familiar attribute (mileage) carries a large, significant valuation ($-\$9,950^{**}$ per 10,000 miles) while most others are imprecisely estimated, and completion times are the shortest of any class except Class 5 (81 seconds across six tasks). Ignoring attributes is known to distort WTP estimates (Hensher et al., 2005), and a standard remedy is to let a latent class absorb non-attending behavior (Scarpa et al., 2009). We therefore retain Class 6 in estimation, where it serves this function, but flag its WTP values as unreliable and exclude them from cross-class comparisons.

4. Conclusion

This paper provides the first WTP estimates for battery health attributes in the U.S. used BEV market, based on a national DCE with 3,072 respondents and a six-class LCCM estimated on 2,916 respondents with complete covariate data. Our LCCM model reveals that consumers can be organized into six segments that differ markedly in their sensitivity to battery quality signals, their price responsiveness, and their market engagement. **Class 1** (22.7%), the largest class, is characterized by the strongest and most uniform range loss aversion (approximately \$2,300 per percentage point across all degradation segments) and meaningful disutility for both refurbishment types; these well-informed consumers are effectively purchasing battery reliability rather than range per se. **Class 2** (20.8%) attends simultaneously to range adequacy, degradation trajectory, and refurbishment condition, and opts out selectively when no presented vehicle meets its standards across all dimensions at once. **Class 3** (16.6%) is the most EV-knowledgeable

segment and places an exceptionally high premium on crossing a minimum viable range threshold, but assigns negligible disutility to refurbishment history. **Class 4** (12.8%) is the most price-sensitive class and is composed of budget-constrained buyers who are strongly committed to making a vehicle choice but largely indifferent to battery quality signals. **Class 5** (8.7%) is defined by systematic opt-out behavior and represents consumers who are largely outside the addressable used BEV market, with low BEV knowledge, high range anxiety, and predominantly ICEV households. **Class 6** (18.5%) is a low-engagement, attribute non-attendance class: its near-zero price sensitivity mechanically inflates WTP ratios, so we flag its estimates as unreliable and withhold substantive interpretation.

These findings carry a direct message for dealers and OEMs. Vehicles with documented performance indicators (mileage, range) and battery health attributes (degradation rate and refurbished history) represent a material pricing premium opportunity with the majority of the market. It is important to consider the development of standardized battery health certification, analogous in structure and function to vehicle history reports. A certification program that provides independently verified SOH readings, degradation history, and remaining warranty coverage could allow dealers to differentiate inventory and command measurable price premiums from the battery-quality-sensitive majority. Critically, cell and pack replacement are penalized at similar magnitudes despite their technical differences, which indicates that consumers do not currently distinguish between high-quality refurbishment and simple repair. This represents a market opportunity for third-party certification services that validate refurbishment quality and communicate it in terms consumers understand.

CRediT authorship contribution statement

Xiatian Iogansen: Conceptualization, Data curation, Methodology, Software, Formal analysis, Writing - Original Draft, Writing - Review & Editing. **Zain Hoda:** Conceptualization, Data curation, Methodology, Writing - Review & Editing. **Christina Gore:** Conceptualization, Data curation, Methodology, Investigation, Writing - Original draft preparation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Joshua D. Kneifel:** Conceptualization, Data curation, Methodology, Investigation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Sindhu Ranganath:** Writing - Original Draft, Writing - Review & Editing. **John P. Helveston:** Conceptualization, Methodology, Writing - Review & Editing, Supervision.

References

- Boudway, I. (Dec. 16, 2020). "Batteries For Electric Cars Speed Toward a Tipping Point". In: *Bloomberg.com*. URL: <https://www.bloomberg.com/news/articles/2020-12-16/electric-cars-are-about-to-be-as-cheap-as-gas-powered-models> (visited on 07/03/2025).
- Chen, C.-F., C. Fu, and Y.-C. Chen (Aug. 2023). "Exploring Tourist Preference for Mobility-as-a-Service (MaaS) – A Latent Class Choice Approach". In: *Transportation Research Part A: Policy and Practice* 174, p. 103750. ISSN: 09658564. DOI: 10.1016/j.tra.2023.103750. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0965856423001702> (visited on 05/18/2026).
- Cox Automotive Inc. (2025). *EV Market Monitor – February 2025*. URL: <https://www.coxautoinc.com/market-insights/ev-market-monitor-february-2025/> (visited on 06/30/2025).
- Fernandes, J. (2023). *US: Understanding Consumer Opinion about Used Electric Cars*. URL: <https://business.yougov.com/content/47834-us-understanding-consumer-opinion-about-used-electric-cars> (visited on 06/30/2025).
- Hensher, D. A., J. Rose, and W. H. Greene (May 1, 2005). "The Implications on Willingness to Pay of Respondents Ignoring Specific Attributes". In: *Transportation* 32.3, pp. 203–222. ISSN: 1572-9435. DOI: 10.1007/s11116-004-7613-8. URL: <https://doi.org/10.1007/s11116-004-7613-8> (visited on 07/09/2026).
- Market Research Future (2025). *United States Used Car Market Trends & Growth By 2034*. URL: <https://www.marketresearchfuture.com/reports/us-used-car-market-21771> (visited on 07/01/2025).
- Scarpa, R., T. J. Gilbride, D. Campbell, and D. A. Hensher (June 1, 2009). "Modelling Attribute Non-Attendance in Choice Experiments for Rural Landscape Valuation". In: *European Review of Agricultural Economics* 36.2, pp. 151–174. ISSN: 0165-1587. DOI: 10.1093/erae/jbp012. URL: <https://doi.org/10.1093/erae/jbp012> (visited on 07/09/2026).



Figure 3: WTP and class profile summary.